

Question 1

Let's break down the properties and the utility function to understand why the answer is **Transitivity and Monotonicity**.

Understanding the Properties

- * **Monotonicity**: A consumer always prefers more of a good to less. In terms of utility functions, this means that if you increase the quantity of one good (holding the other constant), the utility should increase.
- * **Strict Convexity**: A consumer prefers a mix of goods over extreme amounts of a single good. This translates to a utility function where the indifference curves are strictly convex (bowed inward).
- * **Transitivity**: If a consumer prefers good A to good B, and prefers good B to good C, then they must also prefer good A to good C.
- * **Completeness**: A consumer can always compare any two bundles of goods and say they either prefer one, the other, or are indifferent between them.
- * **Continuity**: Small changes in the quantities of goods lead to small changes in utility. This means the indifference curves are continuous (no jumps or breaks).

Analyzing the Utility Function

The utility function $U(x_1, x_2) = 3x_2 - 2x_1$ presents a problem. Let's focus on good 1:

- * **Monotonicity**: As we increase x_1 (the quantity of good 1), the utility *decreases* because of the $-2x_1$ term. This violates monotonicity.
- * **Transitivity**: Consider three bundles:
 - * Bundle A: ($x_1 = 0, x_2 = 1$) $U(A) = 3$

* Bundle B: ($x_1 = 1$, $x_2 = 1$) $U(B) = 1$

* Bundle C: ($x_1 = 2$, $x_2 = 1$) $U(C) = -1$

This function implies that the consumer prefers A to B ($3 > 1$) and B to C ($1 > -1$). However, it also implies that the consumer prefers C to A ($-1 > 3$), violating transitivity.

* **Other Properties:** The utility function does not violate completeness or continuity.

Conclusion

The utility function $U(x_1, x_2) = 3x_2 - 2x_1$ violates **Transitivity and Monotonicity. It's not a valid representation of rational consumer preferences because it suggests a consumer might prefer less of a good to more, and their preferences might not follow a logical order.**

Question 2

Let's break down the concept of completeness and apply it to this scenario.

Completeness Property

The completeness property in preference theory states that for any two items in a set, a consumer can always express a preference:

- * Item A is preferred to item B
- * Item B is preferred to item A
- * The consumer is indifferent between item A and item B

Analyzing the Given Information

We have the following information about preferences:

- * A is preferred to B ($A \succ B$)
- * D is preferred to C ($D \succ C$)
- * C is indifferent to B ($C \sim B$)

Checking for Completeness

We need to check if we can compare ALL possible pairs of items from the set $S = \{A, B, C, D\}$. Let's go through each pair:

- * A vs. B: We know $A \succ B$ (already stated).

* **A vs. C:** We don't have direct information, but we know $A > B$ and $B \sim C$. This implies $A > C$ (transitivity of preferences).

* **A vs. D:** We don't have direct information, and we can't deduce a preference between A and D based on the given information.

* **B vs. C:** We know $C \sim B$ (already stated).

* **B vs. D:** We don't have direct information, and we can't deduce a preference between B and D based on the given information.

* **C vs. D:** We know $D > C$ (already stated).

Conclusion

Since we cannot determine a preference between A and D, or between B and D, the completeness property is **NOT** satisfied.

Answer: No

Question 3

Let's break down the preference information and see if it meets the criteria for transitivity.

Understanding Transitivity

Transitivity in preferences means: If you prefer A to B, and you prefer B to C, then you ***must*** prefer A to C.

Analyzing the Given Preferences

We are given:

* A is preferred to B ($A > B$)

* D is preferred to C ($D > C$)

* C is indifferent to B ($C \sim B$)

* A is indifferent to D ($A \sim D$)

Checking for Transitivity

* **From $A > B$ and $C \sim B$, we can conclude $A > C$.** This is because if A is better than B, and B is as good as C, then A must be better than C.

* **From $D > C$ and $C \sim B$, we can conclude $D > B$.** Similar reasoning applies here.

* **From $A \sim D$ and $D > C$, we can conclude $A > C$.** This is because A is as good as D, and D is better than C, so A must be better than C.

Conclusion

We have no contradictions in our conclusions. Every combination of preferences in the given set satisfies the criteria for transitivity.

Answer: Yes

Question 4

Here's how to find the utility-maximizing bundle for this individual:

Understanding the Problem

* **Utility Function:** $U(x, y) = 2\log x + y$ represents the individual's satisfaction from consuming goods x and y .

* **Budget Constraint:** The individual has a limited income of Rs 50, and the prices of the goods determine how much they can buy. The budget constraint is:

$$x + 2y = 50$$

* **Goal:** We want to find the combination of x and y that maximizes the individual's utility (satisfaction) while staying within their budget.

Solving for the Optimal Bundle

1. Set up the Lagrangian:

We use a Lagrangian to find the optimal combination. The Lagrangian combines the utility function and the budget constraint:

$$L(x, y, \lambda) = 2\log x + y + \lambda(50 - x - 2y)$$

where λ is a Lagrange multiplier.

2. Find the First-Order Conditions (FOCs):

To maximize the utility, we take the partial derivatives of the Lagrangian with respect to x , y , and λ , and set them equal to zero:

$$* \partial L / \partial x = 2/x - \lambda = 0$$

$$* \partial L / \partial y = 1 - 2\lambda = 0$$

$$* \partial L / \partial \lambda = 50 - x - 2y = 0$$

3. **Solve the System of Equations:**

Solving these equations simultaneously, we get:

$$* x = 20$$

$$* y = 15$$

Explanation

The solution ($x = 20$, $y = 15$) represents the utility-maximizing bundle. This means that the individual will get the highest possible level of satisfaction by consuming 20 units of good x and 15 units of good y , given their budget constraint.

Answer: None of the provided options are correct. The utility-maximizing bundle is **$x = 20$ and $y = 15$** .

Question 5

Here's how to find the utility-maximizing consumption bundle:

Understanding the Problem

- * **Utility Function:** The utility function, $U(x,y)$, represents the satisfaction an individual gets from consuming different amounts of goods x and y .
- * **Budget Constraint:** The budget constraint limits how much the individual can spend based on their income and the prices of the goods.
- * **Goal:** The goal is to find the combination of goods (x, y) that maximizes the individual's utility while staying within their budget.

Steps to Solve

1. Set up the Lagrangian:

- * We introduce a Lagrangian function to incorporate the budget constraint into the utility maximization problem.
- * The Lagrangian is: $L = U(x, y) + \lambda(I - P_x \cdot x - P_y \cdot y)$
- * L is the Lagrangian function
- * λ (lambda) is a Lagrange multiplier
- * I is income
- * P_x is the price of good x
- * P_y is the price of good y

2. Take partial derivatives:

* Take partial derivatives of the Lagrangian with respect to x , y , and λ .

$$* \partial L / \partial x = 0.5x^{-0.5}y^{0.6} - 5\lambda = 0$$

$$* \partial L / \partial y = 0.6x^{0.5}y^{-0.4} - 3\lambda = 0$$

$$* \partial L / \partial \lambda = I - P_x x - P_y y = 0$$

3. Solve the system of equations:

* Now you have three equations and three unknowns (x , y , and λ). Solve for x and y .

* From the first two equations, you can get:

$$* x^{0.5}y^{0.6} / (5\lambda) = 1$$

$$* x^{0.5}y^{-0.4} / (3\lambda) = 1$$

$$* \text{Divide these two equations: } (y/5)/(x/3) = 1.2$$

$$* \text{Simplify: } 3y = 6x$$

$$* \text{Therefore, } y = 2x$$

* Substitute this relationship into the third equation (budget constraint):

$$* 100 - 5x - 3(2x) = 0$$

$$* 100 - 11x = 0$$

$$* x = 100/11$$

* Substitute the value of x back into the equation $y = 2x$ to find y :

$$* y = 2 * (100/11) = 200/11$$

Answer

The utility-maximizing consumption bundle is $(x, y) = (100/11, 200/11)$.

Question 6

Let's break down how to find the utility-maximizing consumption bundle.

Understanding the Problem

- * **Utility Function:** The function $U(x,y) = x^{0.5} * y^{0.6}$ represents the individual's satisfaction from consuming goods x and y . Higher values of U mean greater satisfaction.
- * **Budget Constraint:** The individual has a limited income of Rs 150. They must spend this income on goods x and y , considering their prices.
- * **Goal:** We want to find the combination of goods x and y that maximizes the individual's utility, given their budget constraint.

The Solution Approach

1. **Set up the Lagrangian:** We use a mathematical tool called the Lagrangian to solve this problem. The Lagrangian combines the utility function with the budget constraint:

$$L(x, y, \lambda) = U(x, y) + \lambda(I - P_x x - P_y y)$$

Where:

- * λ is a Lagrange multiplier (a variable to help us find the solution)
- * I is the income (Rs 150)
- * P_x is the price of good x (Rs 5)
- * P_y is the price of good y (Rs 3)

2. **Take Partial Derivatives:** We find the partial derivatives of the Lagrangian with respect to x , y , and λ , and set them equal to zero. This is because the maximum value of the Lagrangian occurs when its

derivatives are zero.

$$* \partial L / \partial x = 0.5x^{(-0.5)}y^{0.6} - 5\lambda = 0$$

$$* \partial L / \partial y = 0.6x^{0.5}y^{(-0.4)} - 3\lambda = 0$$

$$* \partial L / \partial \lambda = I - P_x x - P_y y = 0 \text{ (This is our budget constraint)}$$

3. **Solve the System of Equations:** We have three equations and three unknowns (x, y, and λ). Solving this system will give us the values of x and y that maximize utility.
4. **Interpret the Solution:** The values of x and y we find represent the optimal quantities of goods x and y to consume to maximize utility.

Explanation

The Lagrangian helps us find the point where the individual's indifference curve (representing all combinations of goods that give the same level of satisfaction) is tangent to their budget constraint. This point represents the highest level of utility the individual can achieve given their income and prices.

Solving the System

Here's how to solve the system of equations:

1. **Divide the first two equations:** Divide the equation $\partial L / \partial x$ by $\partial L / \partial y$:

$$(0.5x^{(-0.5)}y^{0.6}) / (0.6x^{0.5}y^{(-0.4)}) = (5\lambda) / (3\lambda)$$

Simplifying, we get: $y/x = 10/9$

2. **Solve for y in terms of x:**

$$y = (10/9)x$$

3. Substitute into the budget constraint:

$$150 - 5x - 3[(10/9)x] = 0$$

4. Solve for x:

$$x = 150 / (5 + 10/3) = 150 / (25/3) = 18$$

5. Substitute to find y:

$$y = (10/9) * 18 = 20$$

Therefore, the utility-maximizing consumption bundle is $(x, y) = (18, 20)$.

Important Note: The provided options in your question are incorrect. The correct solution is not within the given choices.

Question 7

Let's break down how to find the demand functions for this utility function.

Understanding the Utility Function

* **Leontief Utility:** This utility function represents a **perfect complements** scenario. The individual gets utility only when consuming goods x and y in a fixed ratio. The "min" function means the individual will consume goods x and y in a way that makes the smallest value of (2x) and (3y) as large as possible.

* **Fixed Ratio:** The ratio of x to y that maximizes utility is 2:3. This means for every 2 units of x consumed, the individual needs 3 units of y.

Deriving the Demand Functions

1. **Budget Constraint:** The individual's spending on x and y cannot exceed their income (m):

$$* P_x * x + P_y * y = m$$

2. **Optimal Consumption Ratio:** As we established, the optimal ratio is $2x = 3y$.

3. **Solving for x and y:**

* **Step 1:** Solve the ratio equation for one variable. Let's solve for y:

$$* y = (2/3)x$$

* **Step 2:** Substitute this expression for y into the budget constraint:

$$* P_x * x + P_y * (2/3)x = m$$

* **Step 3:** Simplify and solve for x:

$$* x * (P_x + (2/3)P_y) = m$$

$$* x = m / (P_x + (2/3)P_y)$$

* **Step 4:** Substitute the value of x back into the equation for y:

$$* y = (2/3) * [m / (P_x + (2/3)P_y)]$$

$$* y = (2m) / (3P_x + 2P_y)$$

Answer

Therefore, the demand functions are:

$$* x = (3m) / (2P_x + 3P_y)$$

$$* y = (2m) / (2P_x + 3P_y)$$

Key Points:

* The demand functions show that the individual's consumption of both goods depends on income (m) and the relative prices of x and y (P_x and P_y).

* The demand for x is higher when the price of y is higher, as the individual will substitute more of x to maintain the desired 2:3 ratio.

* The demand for y is higher when the price of x is higher, for the same reason.

Question 8

Here's how to derive the market demand function from the individual demand functions:

Understanding the Concept

- * **Individual Demand:** Each person's demand function tells us how much they'll buy at different prices.
- * **Market Demand:** The market demand function combines all the individual demand functions to show the total quantity demanded at each price in the entire market.

Steps to Derive the Market Demand

1. **Solve for Quantity:** Rearrange each individual demand function to express quantity (Q) as a function of price (P):

- * Individual 1: $Q = 10 - 2P$

- * Individual 2: $Q = 30 - 3P$

2. **Horizontal Summation:** To find the market demand, we add the quantities demanded by each individual at each price. This is called horizontal summation because we're adding quantities along the horizontal axis of a demand curve.

- * **For any given price (P), add the quantities demanded by each individual:**

- * Total $Q = (10 - 2P) + (30 - 3P)$

- * Total $Q = 40 - 5P$

Important Note: This simple addition only works if the two individuals' demand curves don't intersect within the relevant price range. If they did, we'd need to consider different price segments.

The Answer

The market demand function is: $Q = 40 - 5P$

Explanation:

The market demand function ($Q = 40 - 5P$) tells us that the total quantity demanded in the market will be equal to 40 minus 5 times the price. For example, if the price is \$2, the total quantity demanded will be 30 units ($40 - (5 * 2)$).

Why the other options are incorrect:

* The options with different equations depending on price ranges are incorrect. Since the demand curves don't intersect in the price range we're looking at, the market demand is a single function.

Let me know if you have any more questions!

Question 9

Here's how to determine which demand function is more elastic:

Understanding Elasticity

Price elasticity of demand measures how responsive the quantity demanded of a good is to changes in its price.

- * **Elastic Demand:** If the quantity demanded changes significantly when the price changes, the demand is considered elastic (greater than 1).
- * **Inelastic Demand:** If the quantity demanded changes very little or not at all when the price changes, the demand is considered inelastic (less than 1).
- * **Unit Elastic Demand:** If the quantity demanded changes proportionally to the price change, the demand is unit elastic (equal to 1).

Calculating Elasticity

There are a few ways to calculate elasticity, but the easiest for this problem is to use the following formula:

- * **Price Elasticity of Demand (Ed) = (% Change in Quantity Demanded) / (% Change in Price)**

Since we don't have specific price and quantity changes, we can use a shortcut to compare the elasticities of the two goods. The shortcut relies on the idea that the steeper the demand curve, the less elastic the demand. This is because a steeper curve indicates a smaller change in quantity demanded for a given change in price.

Analyzing the Demand Functions

1. **Good 1:** $4 - P = Q$

- This equation can be rearranged to solve for P: $P = 4 - Q$.

- The coefficient of Q is -1. A coefficient of -1 indicates a steeper slope than the demand curve for good 2.

2. **Good 2:** $3 - 0.4P = Q$

- Rearranging to solve for P: $P = 7.5 - 2.5Q$

- The coefficient of Q is -2.5. A coefficient of -2.5 indicates a flatter slope than the demand curve for good 1.

Conclusion

Since the demand curve for good 2 is flatter, it means that the quantity demanded of good 2 changes more significantly in response to a price change compared to good 1. Therefore, the demand function of good 2 is more elastic.